

Pseudomonad Lanthanide-Switch PQQ Alcohol Oxidation: A Review-Style Synthesis

A commissioned review of the PedS2/PedR2 two-component system coupling lanthanide availability to metal-conditioned periplasmic PQQ alcohol dehydrogenases PedE and PedH.

1. Executive Summary

The **pseudomonad lanthanide switch** is a compact, reusable regulatory-catalytic module that allows a soil/rhizosphere bacterium to select which of two periplasmic pyrroloquinoline-quinone (PQQ) alcohol dehydrogenases (ADHs) it deploys, according to the availability of usable light lanthanides (rare-earth elements, REEs). The module has three linked parts: (i) a **lanthanide-responsive sensor histidine kinase, PedS2**; (ii) a **cognate LuxR-type response regulator, PedR2**, that reads the phosphorylation state set by PedS2; and (iii) two paralogous, functionally overlapping **quinoprotein alcohol dehydrogenases, PedE (Ca²⁺-dependent) and PedH (lanthanide-dependent)**, that oxidize a broad range of alcohols and aldehydes in the periplasm and pass electrons to cytochrome *c*.

The best-supported model (worked out chiefly in *Pseudomonas putida* KT2440) is a phosphorylation-gated inverse switch. In the **absence of usable lanthanides**, PedS2 phosphorylates PedR2; phospho-PedR2 **activates *pedE*** and simultaneously **contributes to repression of *pedH***. When **light lanthanides (La³⁺–Nd³⁺) become available**, PedS2 kinase activity falls — proposed to result from lanthanide perception at its periplasmic region — lowering phospho-PedR2 so that *pedE* transcription declines and *pedH* repression is relieved. Notably, full *pedH* activation additionally requires a second, still-unidentified regulatory input, so the switch is **not symmetric**: de-repression of *pedH* is necessary but not sufficient (Wehrmann et al., 2017 [PMID 28655819]; Wehrmann et al., 2018 [PMID 30158283]).

This system is best understood as a **heterotrophic elaboration of an ancient scaffold** — two-component regulation of a PQQ alcohol dehydrogenase feeding cytochrome *c* — that predates the lanthanide dimension in pseudomonads (ExaD/ExaE and AgmR systems; Görisch 2003

[PMID 12686116]; Vrionis et al. 2002 [PMID 11954793]). It is mechanistically **analogous but not identical** to the methylotroph XoxF/MxaF "lanthanide switch," from which much of the structural and chemical rationale is borrowed (Skovran et al. 2019 [PMID 31166187]; Yu & Chistoserdova 2019 [PMID 31166192]). Adjacent systems — **lanthanide uptake (PedA1A2BC), PQQ biosynthesis, and downstream aldehyde/carbon metabolism** — set the inputs and consume the outputs of the module but lie outside its core. The most important open questions are the molecular identity of the additional *pedH* activator and direct structural proof that PedS2 senses lanthanides.

2. Definition and Biological Boundaries

2.1 What is inside the module

The module comprises exactly four dedicated molecular players and their regulatory relationships:

Component	Identity	Role in the module
PedS2 (e.g., PP_2671 in KT2440)	Sensor histidine kinase	Perceives lanthanide status; sets PedR2 phosphorylation
PedR2 (e.g., PP_2672)	LuxR-type response regulator	Dual activator/repressor; transcriptional output
PedE	Ca ²⁺ -dependent PQQ-ADH	Alcohol oxidation when lanthanides unavailable
PedH	Lanthanide-dependent PQQ-ADH	Alcohol oxidation when lanthanides available

Both enzymes are periplasmic quinoproteins that oxidize a broad substrate range of alcohols (and some aldehydes) and transfer electrons to cytochrome *c*, connecting to the respiratory chain. The regulatory logic — a phospho-relay two-component system driving reciprocal expression of a Ca/Ln enzyme pair — is the defining feature.

2.2 What is adjacent but outside the module

Several closely coupled processes are frequently discussed alongside the switch but are mechanistically distinct and should be treated separately:

- **Lanthanide uptake and trafficking** — e.g., the ABC-type transporter **PedA1A2BC** and lanthanide chaperones such as lanmodulin. Uptake sets the intracellular lanthanide concentration that the switch experiences but is not part of the sensing/output circuit itself (Wehrmann et al. 2019 [PMID 31736923]; Cotruvo 2020 [PMID 32979423]).
- **PQQ biosynthesis** — supplies the essential cofactor shared by both enzymes but is a separate biosynthetic pathway.
- **Downstream aldehyde and carbon metabolism** — e.g., aldehyde dehydrogenases (PedI-type) and the novel PedE/PedH-initiated glycerol-oxidation route into central metabolism. These consume the module's products but do not participate in the metal-conditioned switch (Wehrmann et al. 2020 [PMID 32345644]).

2.3 Competing and overlapping definitions

The term "**lanthanide switch**" was first coined in the methylotrophy literature for the reciprocal regulation of **XoxF (Ln-dependent) versus MxaFI (Ca-dependent) methanol dehydrogenases** (Skovran et al. 2019 [PMID 31166187]; Yu & Chistoserdova 2019 [PMID 31166192]). The pseudomonad PedE/PedH system is the **archetypal non-methylotrophic instance** of the same regulatory logic applied to multi-carbon (non-methanol) alcohol metabolism. Readers should be careful not to conflate:

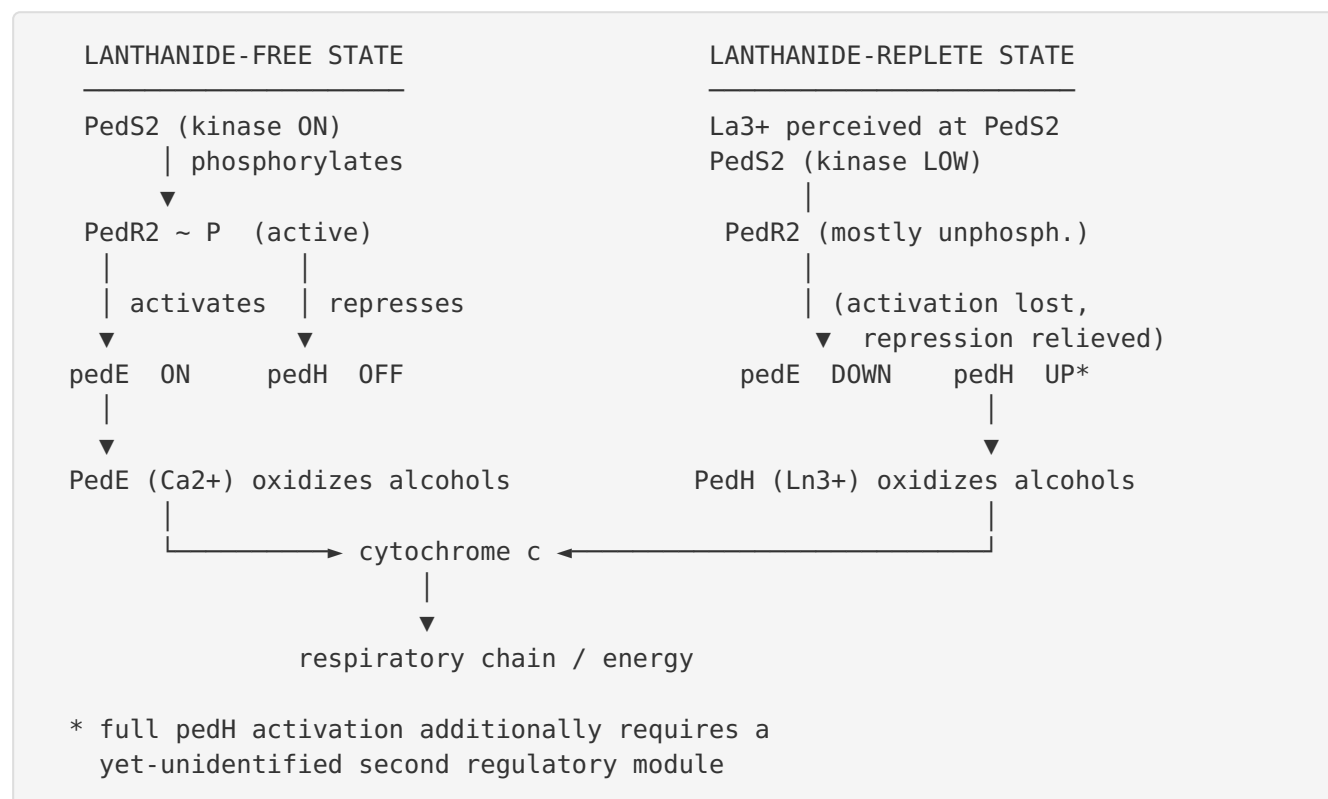
- the **methylotroph methanol-oxidation switch** (XoxF/MxaF), with
- the **pseudomonad broad-alcohol switch** (PedE/PedH, regulated by PedS2/PedR2).

They are homologous in concept and enzyme family but differ in the specific regulators, enzyme architecture details, and physiological substrate.

3. Mechanistic Overview

3.1 The core switch model

The best current model, established primarily through adaptive laboratory evolution, site-directed mutagenesis, reporter fusions, and complementation in *P. putida* KT2440 (Wehrmann et al. 2018 [PMID 30158283]), is as follows:



3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** PedS2-dependent phosphotransfer to PedR2 is the central node. In the lanthanide-free state, phospho-PedR2 is required both to activate *pedE* and to help repress *pedH*.
- **Conditional:** Lanthanide perception is the conditional input that flips the switch. Its downstream consequence — reduced PedS2 kinase activity — is what lowers phospho-PedR2.
- **Accessory / incompletely defined:** Full activation of *pedH* is **not** achieved by de-repression alone; a **second, uncharacterized activator module** is required. This is the single largest mechanistic gap in the model.

3.3 Metal-conditioned catalysis

Once expressed, each enzyme performs the same chemical reaction — PQQ-dependent alcohol oxidation with electron transfer to cytochrome *c* — but uses a different catalytic metal at the PQQ active site. The choice of Ca^{2+} versus Ln^{3+} is not arbitrary: lanthanides are **superior Lewis acids** at the PQQ site. Density-functional calculations on the cerium-dependent methanol dehydrogenase active site show that Ce(III) stabilizes the PQQ LUMO by ~ 0.81 eV relative to calcium, providing an electronic rationale for a catalytic advantage (Bogart et al. 2015 [PMID 25421364]). Cells that can acquire REEs therefore gain access to a kinetically favorable oxidation route.

4. Major Molecular Players and Active Assemblies

4.1 PedS2 — lanthanide-responsive sensor histidine kinase

PedS2 is the input node. In the lanthanide-free state it acts as an active kinase, phosphorylating PedR2. Lanthanide availability lowers its kinase activity, proposed to occur either by **direct binding of Ln^{3+} to the periplasmic region of PedS2** or by an uncharacterized indirect interaction (Wehrmann et al. 2018 [PMID 30158283]). Direct structural or biophysical proof of lanthanide binding by PedS2 has **not yet been reported** and remains a key gap. PedS2 is also a proposed target of **mismetallation**: iron (and to a lesser extent copper and zinc) can raise the minimal Ln^{3+} concentration required to flip the switch by orders of magnitude, consistent with Fe-driven mismetallation of putative La^{3+} -binding proteins such as PedS2 (Wehrmann et al. 2019 [PMID 31736923]).

4.2 PedR2 — LuxR-type response regulator with dual output

PedR2 is a **dual-function** transcriptional regulator: in its phosphorylated form it **activates *pedE*** and **contributes to repression of *pedH***. This makes the switch reciprocal from a single regulator's phosphorylation state (Wehrmann et al. 2018 [PMID 30158283]). Because full *pedH* activation requires an additional module, PedR2 is best described as the master regulator of the Ca-branch and a partial (necessary-but-not-sufficient) contributor to control of the Ln-branch.

4.3 PedE — calcium-dependent PQQ alcohol dehydrogenase

PedE is a Ca^{2+} -dependent quinoprotein ADH that oxidizes a broad range of alcohols/aldehydes in the periplasm and passes electrons to cytochrome *c*. It is the "default" enzyme when usable lanthanides are absent (Wehrmann et al. 2017 [PMID 28655819]).

4.4 PedH — lanthanide-dependent PQQ alcohol dehydrogenase

PedH is the **first lanthanide-dependent PQQ-ADH characterized in a non-methylophilic bacterium**. Purified from *P. putida* KT2440, it requires light lanthanides (La^{3+} – Nd^{3+}) for alcohol-oxidizing activity, whereas its paralog PedE is Ca^{2+} -dependent. The two are **functionally redundant** periplasmic isozymes with overlapping broad substrate ranges (Wehrmann et al. 2017 [PMID 28655819]).

4.5 Structural context of the enzyme pair

PedE and PedH are members of the **QEDH/XoxF-type** family of PQQ-ADHs. Crystallography of the lanthanide methanol dehydrogenase XoxF reveals a **La(III) ion in the active site within a homodimeric fold**, contrasting with the Ca^{2+} -dependent **MxaFI heterotetramer** (Deng et al. 2018 [PMID 30132076]). By homology, PedE/PedH are expected to be XoxF/QEDH-like homodimeric PQQ-ADHs, distinct from the classic MxaFI architecture. These methylophilic enzymes are the best-characterized structural proxies, and inferences to PedE/PedH should be made with that caveat in mind.

Feature	Ca^{2+} branch (PedE / MxaF-like)	Ln^{3+} branch (PedH / XoxF-like)
Catalytic metal	Ca^{2+}	Light Ln^{3+} (La–Nd)
Structural archetype	MxaFI heterotetramer	XoxF homodimer
Electronic effect on PQQ	Baseline	LUMO stabilized ~ 0.81 eV (Ce vs Ca)
Regulatory state favoring it	phospho-PedR2 high	phospho-PedR2 low + extra activator
Electron acceptor	cytochrome <i>c</i>	cytochrome <i>c</i>

5. Evolutionary and Cell-Biological Variation

5.1 An ancient two-component/PQQ-ADH scaffold

Two-component regulation of a PQQ alcohol-oxidation system feeding cytochrome *c* **predates** the lanthanide dimension in pseudomonads. In *Pseudomonas aeruginosa*, transcription of the quinoprotein ethanol dehydrogenase (QEDH/ExaA) is controlled by a two-component system — histidine kinase **ExaD** and response regulator **ExaE** — with cytochrome c_{550} as the essential electron acceptor (Görisch 2003 [PMID 12686116]). *P. putida* KT2440 was already known to carry two PQQ-linked ADH genes and a response regulator (**AgmR**) governing alcohol utilization before the lanthanide role was recognized (Vrionis et al. 2002 [PMID 11954793]). The lanthanide switch is thus best read as a **lineage-specific elaboration** — the addition of metal-sensing input and a Ln-dependent paralog — onto a pre-existing TCS→PQQ-ADH→cytochrome-*c* scaffold.

5.2 The methylotroph connection

The reciprocal Ca/Ln regulatory logic is shared with, and was first defined in, methylotrophs, where the **XoxF (Ln)** / **MxaF (Ca)** methanol dehydrogenase switch is central to one-carbon metabolism (Skovran et al. 2019 [PMID 31166187]; Yu & Chistoserdova 2019 [PMID 31166192]). The recognition that lanthanide-dependent enzymes extend to **multi-carbon metabolism in non-methylotrophs** — with PedE/PedH as the archetype — expanded the scope of lanthanide biochemistry well beyond methylotrophy (Skovran et al. 2019 [PMID 31166187]).

5.3 Conservation and elaboration beyond *Pseudomonas*

The Ca/Ln paralog pair and its inverse regulation are **portable across lineages**. In the plastic-degrading betaproteobacterium *Ideonella sakaiensis*, homologs **IsPedE (Ca²⁺)** and **IsPedH (Pr³⁺)** were identified alongside an additional **IsXoxF (Pr³⁺)** with a larger catalytic pocket that favors bulkier substrates, plus the aldehyde dehydrogenase **IsPedI** — all homologs of *P. putida* KT2440 ethylene-glycol metabolic enzymes. Pr³⁺ negatively regulated IsPedE protein and positively regulated IsPedH and IsXoxF, **reproducing the inverse REE-switch expression pattern in a distinct lineage** (Hachisuka et al. 2022 [PMID 36289066]).

This example also refines the "functional redundancy" claim: the paralogs are **tuned rather than perfectly redundant**. Substrate partitioning — a broad-range PedE, a small-alcohol-preferring PedH, and a bulky-substrate-preferring XoxF — shows the enzyme set is diversified to cover complementary substrate space, with lineage-specific additions (a third XoxF-type enzyme) expanding range.

5.4 Compartmental and physiological states

The enzymes operate in the **periplasm**; the regulators span the membrane (sensor) and cytoplasm (response regulator). The switch is physiologically important under conditions requiring efficient growth on **alcoholic volatiles**, and lanthanum strongly affects growth on **glycerol**, where PedE/PedH initiate a novel oxidation route (to glycerate via glyceraldehyde, then GarK) operating in parallel with the canonical glpFKRD pathway. The cellular La^{3+} response is largely **substrate-specific** rather than global (Wehrmann et al. 2020 [PMID 32345644]).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

- **Sensing precedes output:** PedS2 must set PedR2's phosphorylation state before transcriptional output changes.
- **Cofactor and metal must be available for catalysis:** Both enzymes require PQQ; PedE additionally requires Ca^{2+} and PedH requires a usable light Ln^{3+} . If the appropriate metal is unavailable to the expressed enzyme, oxidation fails regardless of transcription.
- **Uptake gates the input:** Intracellular lanthanide availability — not merely external concentration — determines the switch state, and uptake is gated by the PedA1A2BC transporter (see below).

6.2 Mutually exclusive / conditional relationships

- The **Ca and Ln branches are reciprocally regulated:** high phospho-PedR2 favors PedE and represses PedH, and vice versa. They are not simultaneously maximally expressed.
- *pedH* de-repression is **necessary but not sufficient** for its full expression; the second activator module is a required conditional input.

6.3 Uptake gating and metal competition (failure modes)

REE-dependent growth on 2-phenylethanol at low (nanomolar) Ln^{3+} requires the ABC-type uptake system **PedA1A2BC**. Its loss raises the La^{3+} needed for PedH-dependent growth ~100-fold, **but not** the concentration needed to repress PedE — indicating uptake specifically feeds the enzyme/activation arm (Wehrmann et al. 2019 [PMID 31736923]).

Metal competition is a major failure mode and confounder. Copper, zinc, and especially **iron** raise the minimal Ln^{3+} concentration required for the switch by orders of magnitude, consistent with **Fe-driven mismetallation** of putative La^{3+} -binding proteins such as PedS2 (Wehrmann et al. 2019 [PMID 31736923]). Consequently, the **La^{3+}/Fe ratio** — and, more broadly, medium composition — strongly affects apparent switch thresholds. This is a crucial caveat when comparing quantitative results across studies and organisms.

6.4 Boundary dependencies

The module depends on but does not contain: PQQ biosynthesis (cofactor supply), lanthanide uptake/trafficking (input concentration), and downstream aldehyde/carbon metabolism (product disposal and energy capture). Failures in any of these adjacent systems will impair the module's physiological output without altering its intrinsic switch logic.

7. Controversies and Open Questions

1. **Identity of the second *pedH* activator.** The most important unresolved mechanistic point. Loss of PedR2-mediated repression is insufficient for full *pedH* expression; a distinct, uncharacterized regulatory module is required (Wehrmann et al. 2018 [PMID 30158283]). Its identity, whether it directly senses lanthanides, and how it integrates with PedR2 are unknown.
2. **Direct proof of lanthanide sensing by PedS2.** Lanthanide perception at the PedS2 periplasmic region is **inferred, not structurally demonstrated**. Whether Ln^{3+} binds PedS2 directly or acts through an intermediary remains open (Wehrmann et al. 2018 [PMID 30158283]).

3. **Degree of enzyme redundancy vs specialization.** Early work emphasized functional redundancy of PedE/PedH (Wehrmann et al. 2017 [PMID 28655819]), but cross-lineage data reveal substrate partitioning and additional paralogs (Hachisuka et al. 2022 [PMID 36289066]). How much substrate specialization exists in *P. putida* itself, versus other organisms, is not fully resolved.
4. **Comparability across organisms and assays.** Much of the structural and chemical rationale is imported from **methylobacter** enzymes (XoxF/MxaF) and DFT models (Bogart et al. 2015 [PMID 25421364]; Deng et al. 2018 [PMID 30132076]). Extrapolation to PedE/PedH is reasonable by homology but not directly proven at the structural level. Quantitative switch thresholds are strongly medium-dependent (iron, other metals), complicating cross-study comparison (Wehrmann et al. 2019 [PMID 31736923]).
5. **Generality of the regulatory architecture.** Whether the PedS2/PedR2 architecture (as opposed to the methylobacter MxbD/MxcQ-type systems) is the dominant mode of lanthanide-switch control in non-methylobacters broadly, or is one of several convergent solutions, is not yet established.

8. Limitations and Knowledge Gaps

This synthesis rests heavily on a single, deeply studied model organism (*P. putida* KT2440) and a small number of laboratories. Several structural claims about PedE/PedH are **inferred from methylobacter homologs** (XoxF/MxaF) rather than proven for the Ped enzymes themselves. The lanthanide-sensing mechanism of PedS2 is **hypothetical at the molecular level**, and the second *pedH* activator is **entirely uncharacterized**. Quantitative parameters (switch thresholds, effective La^{3+} concentrations) are **highly medium-dependent** — particularly sensitive to iron and other divalent/trivalent metals — so numbers are not directly transferable between studies. Cross-lineage conservation is supported by one betaproteobacterial example (*I. sakaiensis*); broader phylogenetic sampling is needed to know how general the PedS2/PedR2 architecture is.

9. Proposed Follow-up Experiments and Actions

1. **Identify the second *pedH* activator** via transposon/CRISPRi screens for mutants that fail to induce *pedH* under lanthanide-replete conditions despite loss of PedR2 repression.
 2. **Biophysically test lanthanide binding to PedS2** — isolate the periplasmic domain and measure direct Ln^{3+} binding (ITC, spectroscopy) versus indirect models, including Fe^{3+} competition to test the mismetallation hypothesis.
 3. **Determine PedE and PedH crystal or cryo-EM structures** to confirm the predicted XoxF-like homodimeric architecture and the metal-coordination geometry, replacing homology inference with direct evidence.
 4. **Systematically map substrate specialization** of PedE vs PedH in *P. putida* under matched conditions to quantify the degree of redundancy versus tuning.
 5. **Survey the PedS2/PedR2 architecture across diverse bacteria** to establish whether it is the dominant non-methylotroph lanthanide-switch solution or one of several convergent designs.
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10. Key References

Citations are drawn from the verified evidence base assembled during this investigation. PMIDs link to PubMed.

- Wehrmann M, et al. *Functional Role of Lanthanides in Enzymatic Activity and Transcriptional Regulation of PQQ-Dependent Alcohol Dehydrogenases in Pseudomonas putida KT2440*. mBio (2017). [PMID: 28655819](#) — First characterization of a lanthanide-dependent PQQ-ADH (PedH) in a non-methylotroph; establishes PedE (Ca^{2+}) / PedH (Ln^{3+}) as functionally redundant periplasmic isozymes.
- Wehrmann M, et al. *The PedS2/PedR2 Two-Component System Is Crucial for the Rare Earth Element Switch in Pseudomonas putida KT2440*. mSphere (2018). [PMID: 30158283](#) — Defines the core switch: phospho-PedR2 activates *pedE* and represses *pedH*; La^{3+} lowers PedS2 kinase activity; full *pedH* activation needs an additional unknown module.
- Wehrmann M, et al. *Lanthanide uptake (Peda1A2BC) and iron cross-talk in the REE switch*. Front. Microbiol. (2019). [PMID: 31736923](#) — ABC transporter gates lanthanide input; iron mismetallation of PedS2 as a failure mode.

- Wehrmann M, et al. *The Cellular Response to Lanthanum Is Substrate Specific and Reveals a Novel Route for Glycerol Metabolism in Pseudomonas putida KT2440*. mBio (2020). PMID: 32345644 — Physiological importance for growth on alcoholic volatiles; PedE/PedH-initiated glycerol route.
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- Yu Z, Chistoserdova L. *Systems Biology Meets Enzymology: Communal Metabolism of Methane and the Role of Lanthanides*. (2019). PMID: 31166192 — Lanthanide switch regulating alternative methanol dehydrogenases.
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- Deng YW, et al. *Crystal structure of the lanthanide methanol dehydrogenase XoxF*. (2018). PMID: 30132076 — La(III) in a homodimeric fold vs the Ca^{2+} MxaFI heterotetramer; structural template for PedE/PedH.
- Cotruvo JA Jr. *The biochemistry of lanthanide acquisition, trafficking, and utilization*. (2020). PMID: 32979423 — Context for lanthanide uptake/trafficking (adjacent system).

Prepared as a commissioned review synthesis. Uncertainty is flagged throughout; claims are attributed to the specific studies that support them, and inferences from methylo troph systems to the pseudomonad enzymes are identified as such.
